

SIDDHANT TANTARPALE

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PROFILE SUMMARY

Second-year B.Sc. AI/ML student with hands-on experience in PyTorch, FastAPI, and production ML deployment. Built and shipped 3 end-to-end AI projects spanning medical imaging, deepfake detection, and LLM-powered SaaS. Comfortable working across the full ML stack model training, optimization, and cloud deployment.

EDUCATION

Bachelor of Science — Artificial Intelligence & Machine Learning

Dr. Homi Bhabha State University | Mumbai, India | 2025 – Present

- Relevant Coursework: Machine Learning Algorithms, Deep Learning, Computer Vision, Data Structures & Algorithms, Statistics for AI, Cloud Computing

TECHNICAL SKILLS

Languages: Python, Go, C, SQL, HTML.

Frameworks & Libraries: PyTorch, FastAPI, NumPy, Pandas, scikit-learn, React.js, Next.js, Node.js

AI / ML: Deep Learning, CNNs, Transfer Learning, Grad-CAM, LLM Inference, Prompt Engineering, Deepfake Detection

Cloud & DevOps: AWS, Docker, HuggingFace Spaces, Vercel, Supabase, PostgreSQL

Tools & Platforms: CUDA, Git, HTML5, Jupyter

PROJECTS

OmniMed — Unified Diagnostic Suite | *Personal Project* | 2026

- Engineered a multi-modal medical imaging platform supporting Neurology (MRI), Orthopaedics (X-ray), and Dermatology (Skin) using PyTorch and Next.js, enabling a single unified clinical inference interface.
- Implemented a Singleton Model-Loading architecture that reduced peak VRAM usage by 57%, enabling high-fidelity inference on standard 6 GB consumer GPUs.
- Eliminated OOM crashes through dynamic CUDA cache management, ensuring stable concurrent multi-model inference across specialties.
- Integrated Grad-CAM visual verification to deliver explainable AI heatmaps, improving clinical transparency and supporting physician decision-making.

Deepfake Detector | *Personal Project* | 2026

- Fine-tuned an EfficientNet-based binary classifier to detect GAN-manipulated video frames, achieving reliable classification of synthetic vs. genuine media on image and video inputs.
- Built an end-to-end inference pipeline in Python that processes video frame-by-frame, extracting frequency-domain features to surface GAN artifacts invisible to the human eye.
- Designed a modular analysis architecture in TypeScript separating ingestion, feature extraction, and classification, enabling easy swapping of backbone models.

SIDDWRITES — AI Text Humanization SaaS | *Personal Project* | 2026

- Engineered a production LLM inference API using FastAPI and LLaMA 3.3 70B, delivering sub-3-second response times via a two-pass pipeline with atomic PostgreSQL RPC batching.
- Deployed a Dockerized FastAPI backend on HuggingFace Spaces with a Vanilla JS frontend on Vercel; integrated Supabase auth, RLS-enforced data access, and usage-tracked free/premium tiers.
- Built a multi-layer prompt injection sanitization pipeline to ensure input integrity and output reliability at scale.
- Implemented dynamic burstiness and perplexity adjustment at inference time to produce stylistically natural output from large language models.

AREAS OF INTEREST

Artificial Intelligence & Machine Learning • Explainable AI (XAI) • Computer Vision • LLM Engineering • Cloud Platforms & MLOps • Algorithm Optimization